

Exercise 12 - $\mathcal{H}_2$ - and $\mathcal{H}_\infty$ -control

Problem 12.1: Hamiltonians and AREs

Consider the RICCATI equation

$$\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \mathbf{X} + \mathbf{X} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} + \mathbf{X} \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \mathbf{X} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{0}.$$

- Construct the Hamiltonian matrix $\mathbf{H}$.
- Determine the eigenvalues and eigenvectors of $\mathbf{H}$.
- Determine the 2-dimensional invariant subspace $\chi_-(\mathbf{H})$ and compute the stable solution of the RICCATI equation.
- Verify your solution of the RICCATI equation with MATLAB.

Solution:

- The RICCATI equation can be rewritten to

$$\mathbf{A}^T \mathbf{X} + \mathbf{X} \mathbf{A} + \mathbf{X} \mathbf{R} \mathbf{X} + \mathbf{Q} = \mathbf{0}$$

with

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The corresponding 4×4 Hamiltonian matrix

$$\mathbf{H} = \begin{bmatrix} \mathbf{A} & \mathbf{R} \\ -\mathbf{Q} & -\mathbf{A}^T \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 & 0 \\ -1 & -1 & 0 & -2 \\ -1 & 0 & -1 & 1 \\ 0 & -1 & 1 & 1 \end{bmatrix}.$$

- The eigenvalues are $\lambda_{1,2} = 2$, $\lambda_{3,4} = -2$ and the corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} 0.9045 \\ -0.3015 \\ -0.3015 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0.2825 \\ -0.5887 \\ 0.1531 \\ 0.7418 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0.4886 \\ -0.1388 \\ 0.8023 \\ -0.3137 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_4 = \begin{bmatrix} 0.5774 \\ 0.5774 \\ 0.5774 \\ 0 \end{bmatrix}.$$

- The eigenvectors $\mathbf{v}_3$ and $\mathbf{v}_4$ are a basis for the stable subspace $\chi_-(\mathbf{H})$, hence

$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix} = [\mathbf{v}_3 \quad \mathbf{v}_4] = \begin{bmatrix} 0.4886 & 0.5774 \\ -0.1388 & 0.5774 \\ 0.8023 & 0.5774 \\ -0.3137 & 0 \end{bmatrix}.$$

The stable solution of the algebraic matrix RICCATI equation is given by

$$\mathbf{X} = \mathbf{X}_2 \mathbf{X}_1^{-1} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}.$$

- (d) We solve the more general RICCATI equation (with a different definition of $\mathbf{R}$) with MATLAB:

```
%% Solution of the Riccati equation
A = [ 1 -1; -1 -1 ];
B = eye( 2 );
Q = eye( 2 );
R = diag( [ 1/2 1/2 ] ); % Caution: Mind MATLAB's definition of R
X = icare( A, B, Q, R )
```

The solution is

$$\mathbf{X} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}.$$

Problem 12.2: $\mathcal{H}_\infty$ -control

Consider the control system

$$\dot{x} = -x + w$$

$$y = 2x - w$$

$$z = x - u.$$

Find a constant controller

$$u = Ky$$

that minimises the $\mathcal{H}_\infty$ norm of the transfer function from w to z . Determine the internally stabilising controller K that satisfies

$$\int_0^\infty \|z(t)\|^2 dt \leq \gamma^2 \int_0^\infty \|w(t)\|^2 dt$$

with the lowest possible constant $\gamma > 0$.

Solution:

Applying the LAPLACE transform on the system equations yields

$$X(s) = \frac{W(s)}{s+1}$$

and hence

$$Z(s) = \frac{W(s)}{s+1} - \frac{2KW(s)}{s+1} + \frac{KW(s)(s+1)}{s+1} = W(s) \frac{Ks - K + 1}{s+1}.$$

The transfer function from $W(s)$ to $Z(s)$ is expanded into partial fractions

$$T_{zw}(s) = \frac{Ks - K + 1}{s+1} = \frac{1}{s+1} - \frac{K(1-s)}{s+1}.$$

The inequality from the problem statement can be rewritten in terms of the $\mathcal{H}_\infty$ -norm:

$$\|T_{zw}(s)\|_\infty^2 \leq \gamma^2 \quad \text{or} \quad \|T_{zw}(s)\|_\infty \leq \gamma.$$

We determine the optimal gain K from the first derivative

$$\frac{dT_{zw}(s)}{ds} = \frac{d}{ds} \left[\frac{Ks - K + 1}{s+1} \right] = \frac{K(s+1) - (Ks - K + 1)}{(s+1)^2} \stackrel{!}{=} 0.$$

Rearranging yields $K = \frac{1}{2}$.

Problem 12.3: $\mathcal{H}_2$ - and $\mathcal{H}_\infty$ -control

Consider the generalised plant in state-space realisation

$$\mathbf{P}(s) \stackrel{s}{=} \begin{bmatrix} \mathbf{A} & \mathbf{B}_1 & \mathbf{B}_2 \\ \mathbf{C}_1 & D_{11} & D_{12} \\ \mathbf{C}_2 & D_{21} & D_{22} \end{bmatrix}$$

$$\text{with } \mathbf{A} = \begin{bmatrix} -5 & 2 & -4 \\ 0 & -3 & 0 \\ 0 & 7 & -1 \end{bmatrix}, \quad \mathbf{B}_1 = \begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}, \quad \mathbf{B}_2 = \begin{bmatrix} 6 \\ 8 \\ -5 \end{bmatrix},$$

$$\mathbf{C}_1 = \begin{bmatrix} -2 & 9 & 4 \end{bmatrix}, \quad \mathbf{C}_2 = \begin{bmatrix} 6 & 3 & -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}.$$

Use MATLAB to solve the following tasks:

- Design an $\mathcal{H}_2$ -optimal controller and plot the step response of the closed-loop system.
- Design an $\mathcal{H}_\infty$ -optimal controller with $\bar{\gamma} = 100$, $\underline{\gamma} = 0$ and an error tolerance of 1%. Plot the step response of the closed-loop system.

Solution:

We define the system in MATLAB and check if all assumptions (except A3 and A4) are satisfied:

```
% System definition and prerequisites

% System definition
A = [ -5 2 -4;
      0 -3 0;
      0 7 -1 ];
B1 = [ 7; -3; 1 ];
B2 = [ 6; 8; -5 ];
C1 = [ -2 9 4 ];
C2 = [ 6 3 -1 ];
D11 = 0;
D12 = 1;
D21 = 2;
D22 = 0;
P = ss( A, [ B1 B2 ], [ C1; C2 ], [ D11 D12; D21 D22 ] );
G = ss( A, B2, C2, D22 );

% Prerequisites
disp( rank( ctrb( G ) ) == min( size( ctrb( G ) ) ) ) )
disp( rank( obsv( G ) ) == min( size( obsv( G ) ) ) ) )
disp( rank( D12 ) == min( size( D12 ) ) )
disp( rank( D21 ) == min( size( D21 ) ) )
disp( D11 == 0 )
disp( D22 == 0 )
```

All assumptions are satisfied, hence we can design $\mathcal{H}_2$ - and $\mathcal{H}_\infty$ -optimal controllers.

- We design the $\mathcal{H}_2$ -optimal controller with MATLAB:

```
% H2-controller design
```

```
% Controller design
Kh2 = -h2syn( P, 1, 1 );

% Step response
step( feedback( G * Kh2, 1 ) );
```

The transfer function of the controller is

$$K(s) = \frac{-18.5s^2 - 125s - 334.5}{s^3 + 65s^2 + 2275s + 9665}.$$

(b) The $\mathcal{H}_\infty$ -optimal design is also done in MATLAB:

```
%% Hinf-controller design

% Controller design
Khinf = - hinfsyn(P,1,1,[0,100]);

% Step response
step( feedback( G * Khinf, 1 ) );
```

It yields the same controller (for this example system):

$$K(s) = \frac{-18.5s^2 - 125s - 334.5}{s^3 + 65s^2 + 2275s + 9665}.$$

Problem 12.4: $\mathcal{H}_2$ - and $\mathcal{H}_\infty$ -control

Consider the single input single output plant

$$G(s) = \frac{s - 10}{(s + 10)(s + 1)}$$

and the weighting functions

$$W_e(s) = \frac{1}{s + 0.001} \quad \text{and} \quad W_u(s) = \frac{s + 2}{s + 10}.$$

Use MATLAB to solve the following tasks:

- Design an $\mathcal{H}_2$ - and an $\mathcal{H}_\infty$ -controller and compare the step responses.
- Compare the GM and PM of the controllers.
- Introduce multiplicative uncertainty $\|\Delta(s)\|_\infty \leq 1$ to the plant $G(s)$ with a lead weighting function

$$w_m(s) = 1.88 \frac{s + 1}{s + 2.5}$$

and redesign the $\mathcal{H}_\infty$ -controller.

- Calculate the robust stability margins of the uncertain system with the $\mathcal{H}_2$ - and the $\mathcal{H}_\infty$ -controller.

Solution:

We start with the definition of the system and the weighting functions:

```
%% System definition
s = tf( 's' );
G = (s - 10) / ( (s + 10) * (s + 1) );

% Weighting functions
We = 1 / (s + 0.001);
Wu = (s + 2) / (s + 10);
```

- (a) We design the controllers in MATLAB using a different way (to Problem 3) of constructing the general control configuration:

```
%% H2 and Hinf design

% Construct augmented plant
P = augw( G, We, Wu, [] );

% H2-controller
Kh2 = h2syn( P );
Lh2 = G * Kh2;

% Hinf-controller
Khinf = hinfsyn( P );
Lhinf = G * Khinf;

% Step responses
step( feedback( Lh2, 1 ) );
hold on;
step( feedback( Lhinf, 1 ) );
legend( 'H2 design', 'Hinf design' );
```

This yields

$$K_2 = \frac{-0.99s^3 - 20.98s^2 - 119.9s - 99.89}{s^4 + 15.06s^3 + 61.91s^2 + 113.1s + 0.11}$$

and

$$K_\infty = \frac{-116.6s^3 - 2448s^2 - 13990s - 11660}{s^4 + 169.5s^3 + 2442s^2 + 8473s + 8.47}.$$

- (b) Let's check the margins:

```
%% Check design margins
[ GMh2, PMh2 ] = margin( Lh2 );
[ GMhinf, PMhinf ] = margin( Lhinf );
```

Controller	GM	PM [°]
K_2	3.80	62.19
K_∞	6.84	68.08

- (c) The introduction of uncertainty yields the family of plants

$$G_p(s) = G(s) (1 + w_m(s)\Delta(s))$$

which is incorporated to the $\mathcal{H}_\infty$ design as follows:

```
%% Hinf design with uncertainty

% Uncertainty definition
Wm = 1.88 * (s + 1) / (s + 2.5);
```

```
Delta = ureal( 'Delta', 0 );
Gp    = G * ( 1 + Wm * Delta );

% Hinf-controller
K     = mixsyn( G, We, Wu, Wm ); % Earlier, Matlab was not able to
% handle uncertainties with Hinf
K2    = mixsyn(Gp,We,Wu);
bode(K,K2);

Tinf  = feedback( Gp * K, 1 );
T2    = feedback( Gp * Kh2, 1 );
step(T2, Tinf );
```

- (d) Robust stability means, that all instances of $G_p(s)$ are stable for all $\|\Delta(s)\|_\infty \leq 1$. We can check this numerically with MATLAB:

```
%% Check robust stability
% H2 design
[ stabMarg2, destabUnc2] = robstab( T2 )

% Hinf design
[ stabMargInf, destabUncInf] = robstab( Tinf )
```

Both closed-loop systems are robustly stable with impressive margins.